PWR_FLAG
GND
PWR_FLAG
+5V

STEP DOWNS

U3
TPS7A2018PDBVR
+5V
IN
EN
GND
OUT
+1V8
C3
1uF
C4
1uF
GND
GND
GND

CLOCK BUFFER

U6A
74LVC125
PDM_CLK_XMOS
2
3
R4
22R
PDM_CLK_Q4
GND

U6B
74LVC125
PDM_CLK_XMOS
5
6
R3
22R
PDM_CLK_Q3
GND

U6C
74LVC125
PDM_CLK_XMOS
9
8
R2
22R
PDM_CLK_Q2
GND

U6D
74LVC125
PDM_CLK_XMOS
12
11
R1
22R
PDM_CLK_Q1
GND

U6E
74LVC125
+1V8
FB11
680R
C2
1uF
C1
100nF
14
VCC
GND
7
GND
GND
GND

Power LED
D2
LED
GND
R5
470R
+5V

+1V8
C5
10uF
C6
4.7uF
C7
4.7uF
C8
4.7uF
C9
4.7uF
GND

+5V
+5V
PDM_CLK_XMOS
PDM_DATA_0
PDM_DATA_1
PDM_DATA_2
PDM_DATA_3
PDM_DATA_4
PDM_DATA_5
PDM_DATA_6
PDM_DATA_7
J19
Conn_01x24
GND

Microphone

Pair0
1V8
CLK
DATA
GND
+1V8
PDM_CLK_Q4
PDM_DATA_7
GND
File: mic-pair.kicad_sch

Pair1
1V8
CLK
DATA
GND
+1V8
PDM_CLK_Q1
PDM_DATA_0
GND
File: mic-pair.kicad_sch

Pair7
1V8
CLK
DATA
GND
+1V8
PDM_CLK_Q4
PDM_DATA_6
GND
File: mic-pair.kicad_sch

Pair2
1V8
CLK
DATA
GND
+1V8
PDM_CLK_Q1
PDM_DATA_1
GND
File: mic-pair.kicad_sch

Pair6
1V8
CLK
DATA
GND
+1V8
PDM_CLK_Q3
PDM_DATA_5
GND
File: mic-pair.kicad_sch

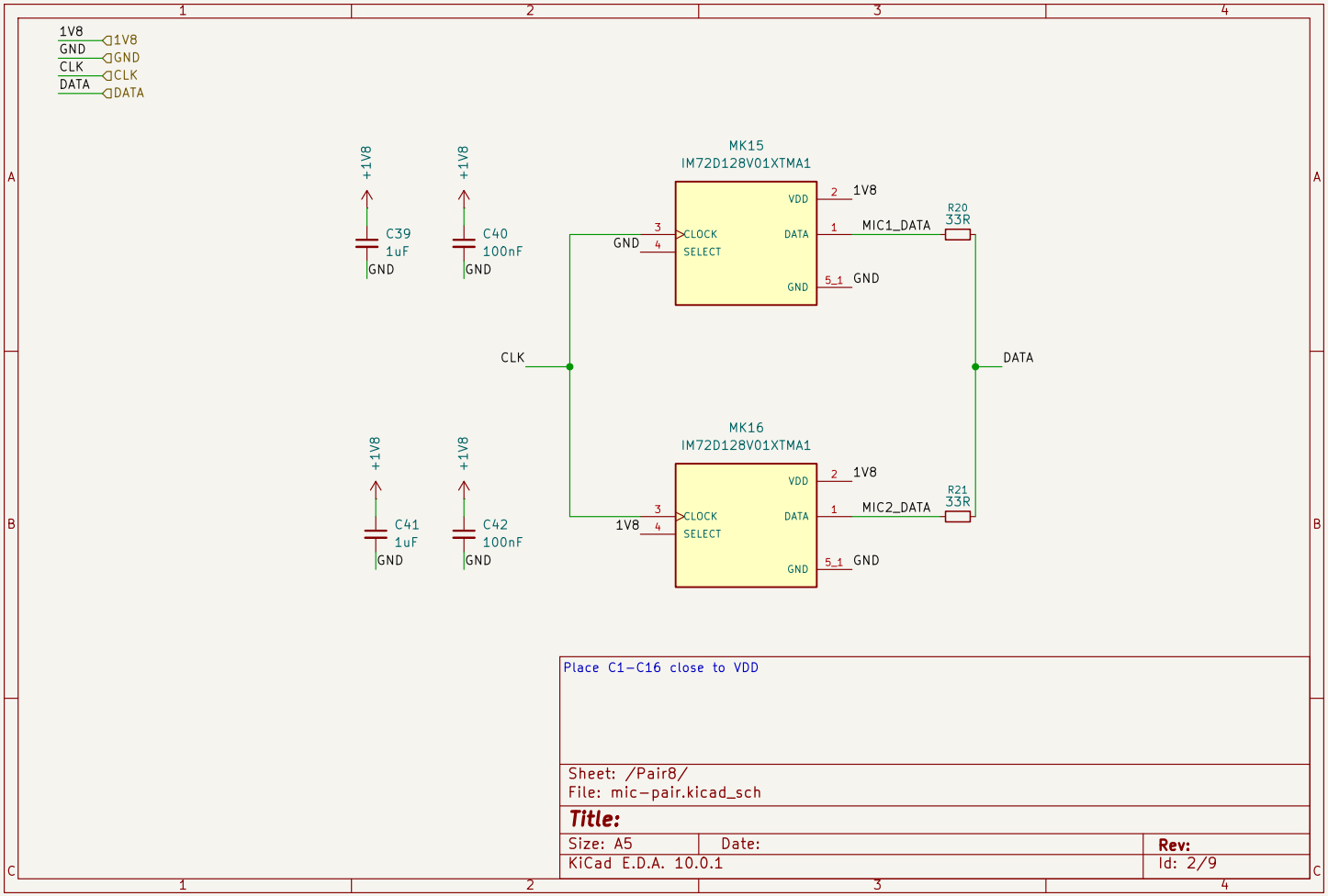
Pair3
1V8
CLK
DATA
GND
+1V8
PDM_CLK_Q2
PDM_DATA_2
GND
File: mic-pair.kicad_sch

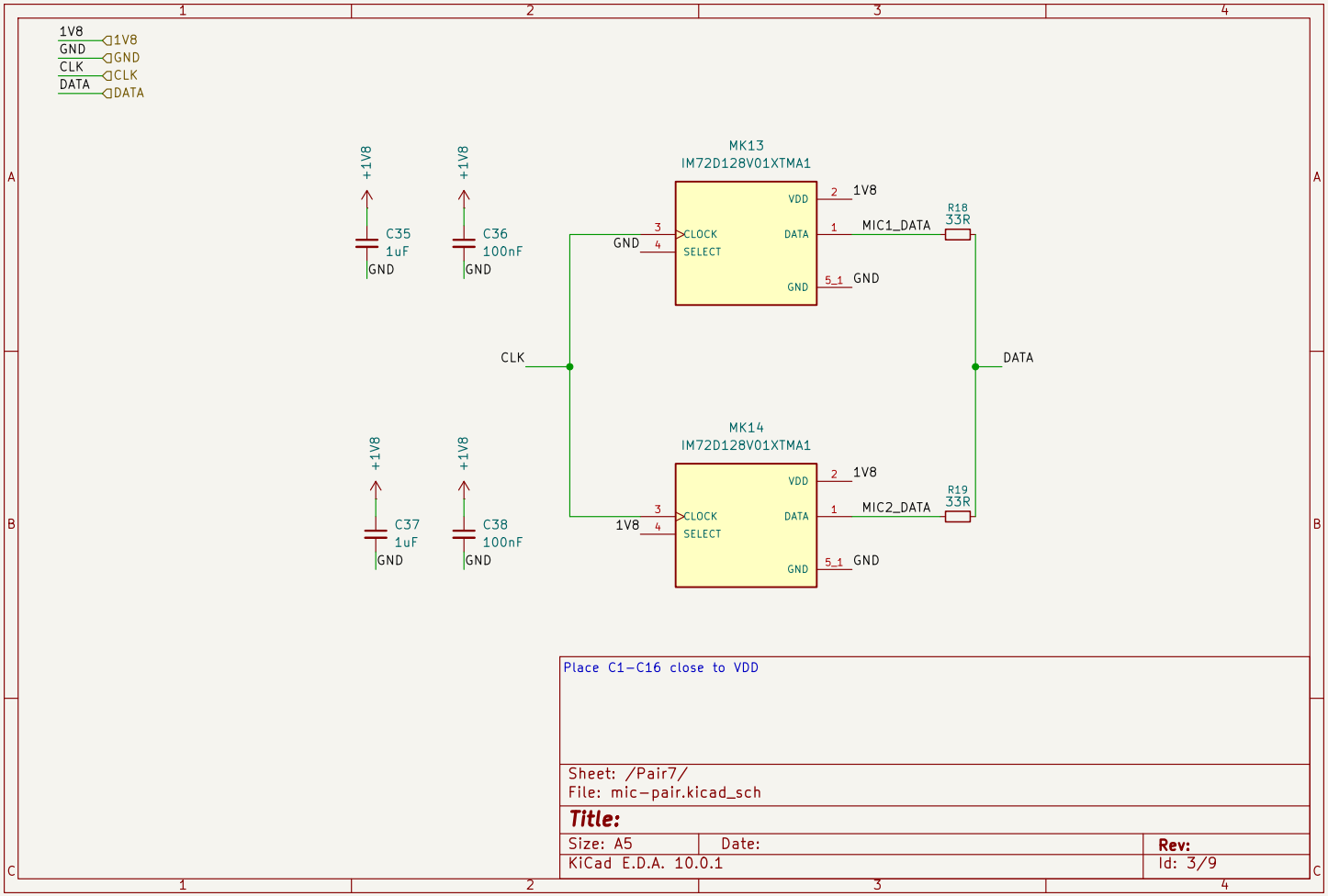
Pair5
1V8
CLK
DATA
GND
+1V8
PDM_CLK_Q3
PDM_DATA_4
GND
File: mic-pair.kicad_sch

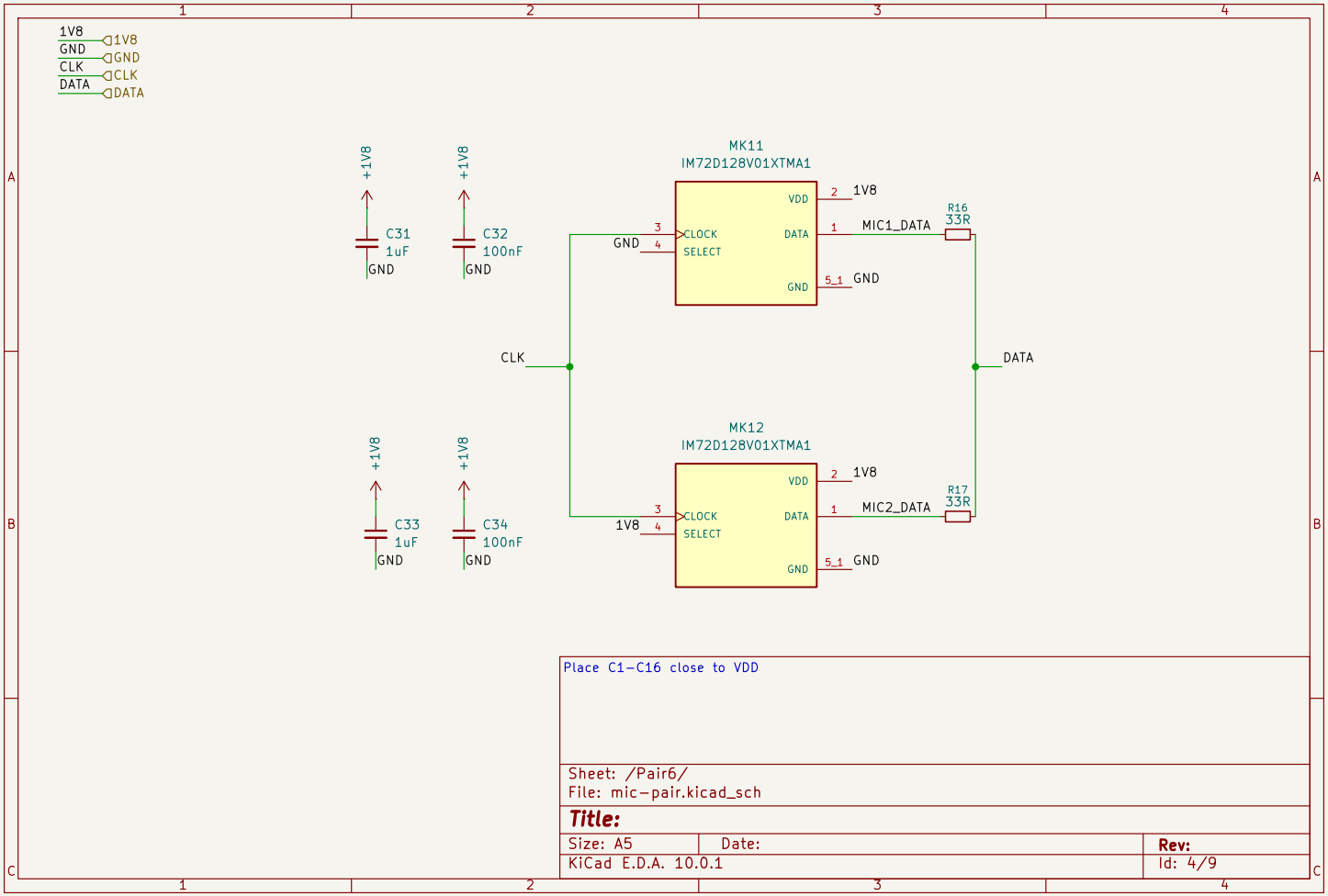
Pair4
1V8
CLK
DATA
GND
+1V8
PDM_CLK_Q2
PDM_DATA_3
GND
File: mic-pair.kicad_sch

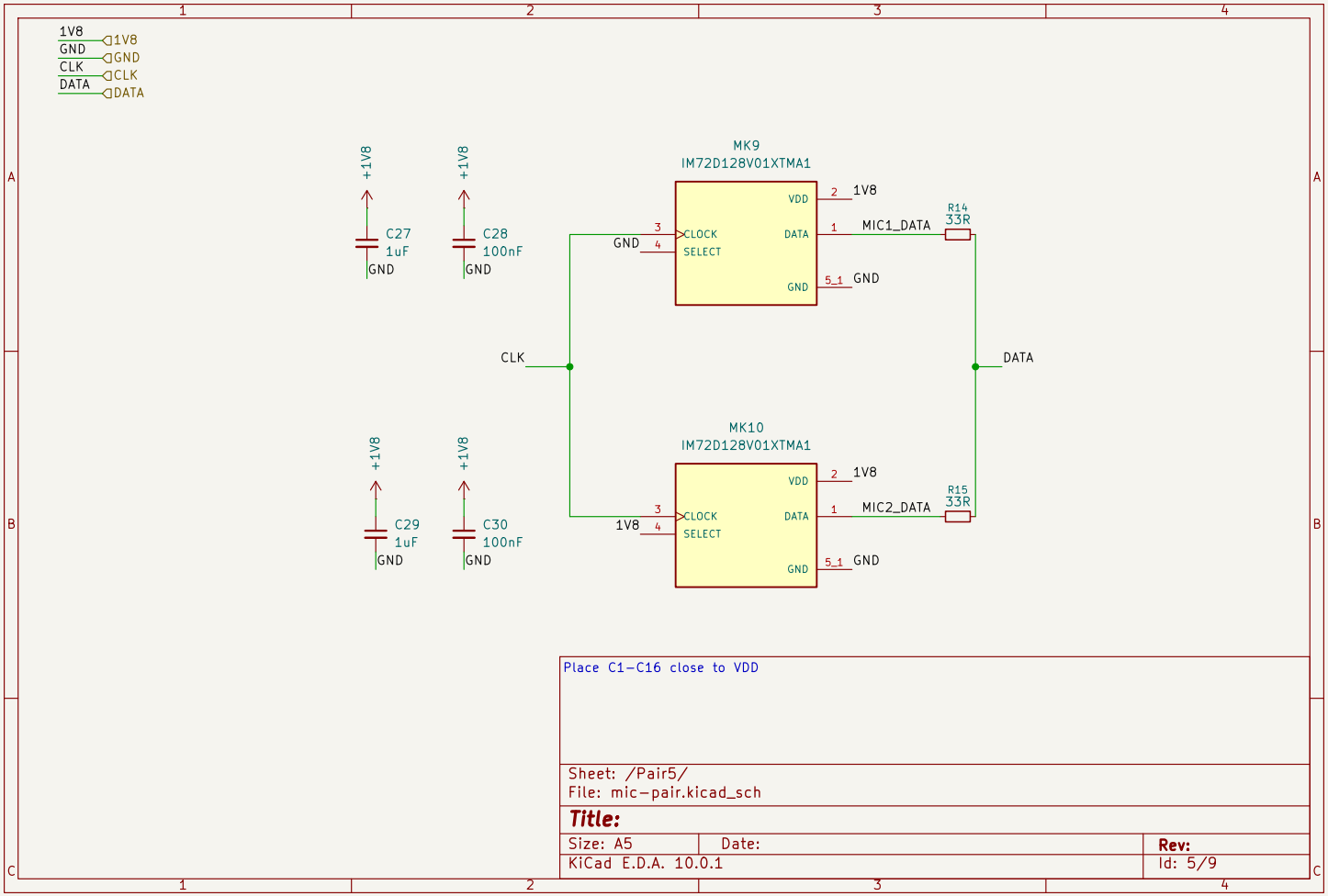
Mounting holes
H1
MountingHole_Pad
H2
MountingHole_Pad
H3
MountingHole_Pad
H4
MountingHole_Pad
H5
MountingHole_Pad
H7
MountingHole_Pad
H6
MountingHole_Pad
H8
MountingHole_Pad
GND

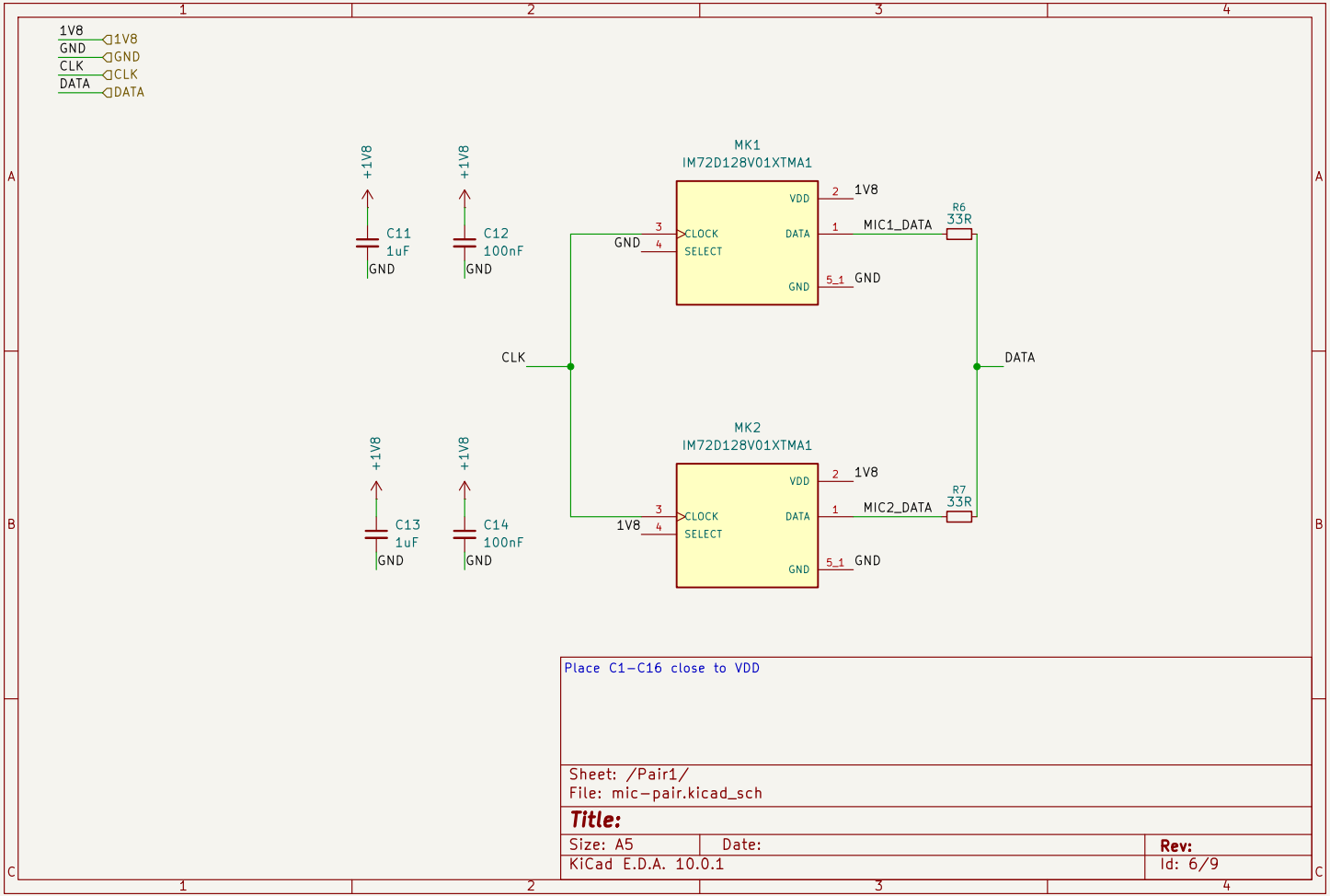
Sheet: /
File: whistler-mic-array.kicad_sch
Title:
Size: A1
Date:
Rev:
KiCad E.D.A. 10.0.1
Id: 1/9

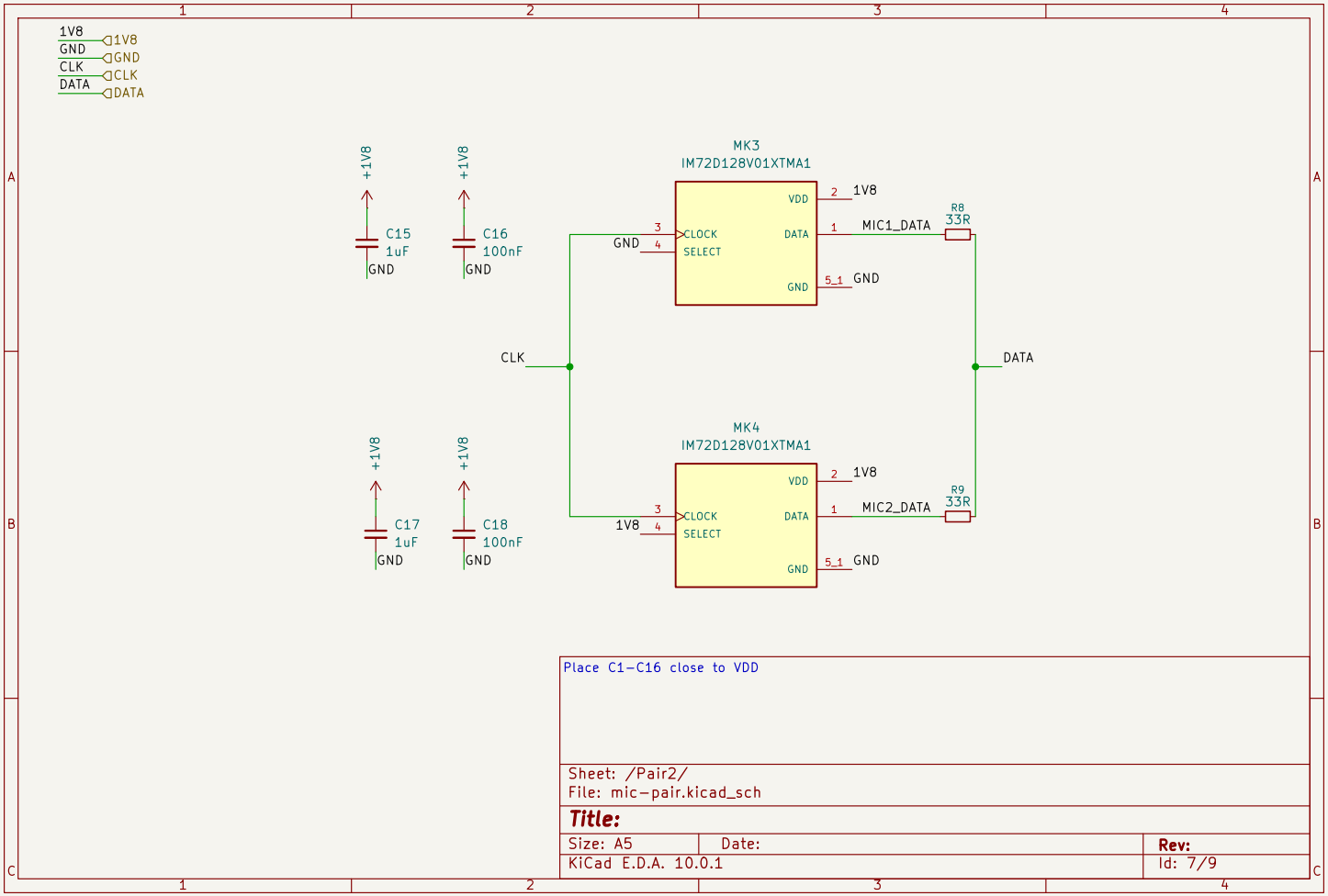

1V8
GND
CLK
DATA
1V8
GND
CLK
DATA
+1V8
C39
1uF
GND
+1V8
C40
100nF
GND
MK15
IM72D128V01XTMA1
VDD
2
1V8
CLOCK
3
GND
SELECT
4
DATA
1
MIC1_DATA
R20
33R
GND
5_1
GND
CLK
DATA
MK16
IM72D128V01XTMA1
VDD
2
1V8
CLOCK
3
1V8
SELECT
4
DATA
1
MIC2_DATA
R21
33R
GND
5_1
GND
+1V8
C41
1uF
GND
+1V8
C42
100nF
GND
Place C1–C16 close to VDD
Sheet: /Pair8/
File: mic–pair.kicad_sch
Title:
Size: A5
Date:
Rev:
KiCad E.D.A. 10.0.1
Id: 2/9

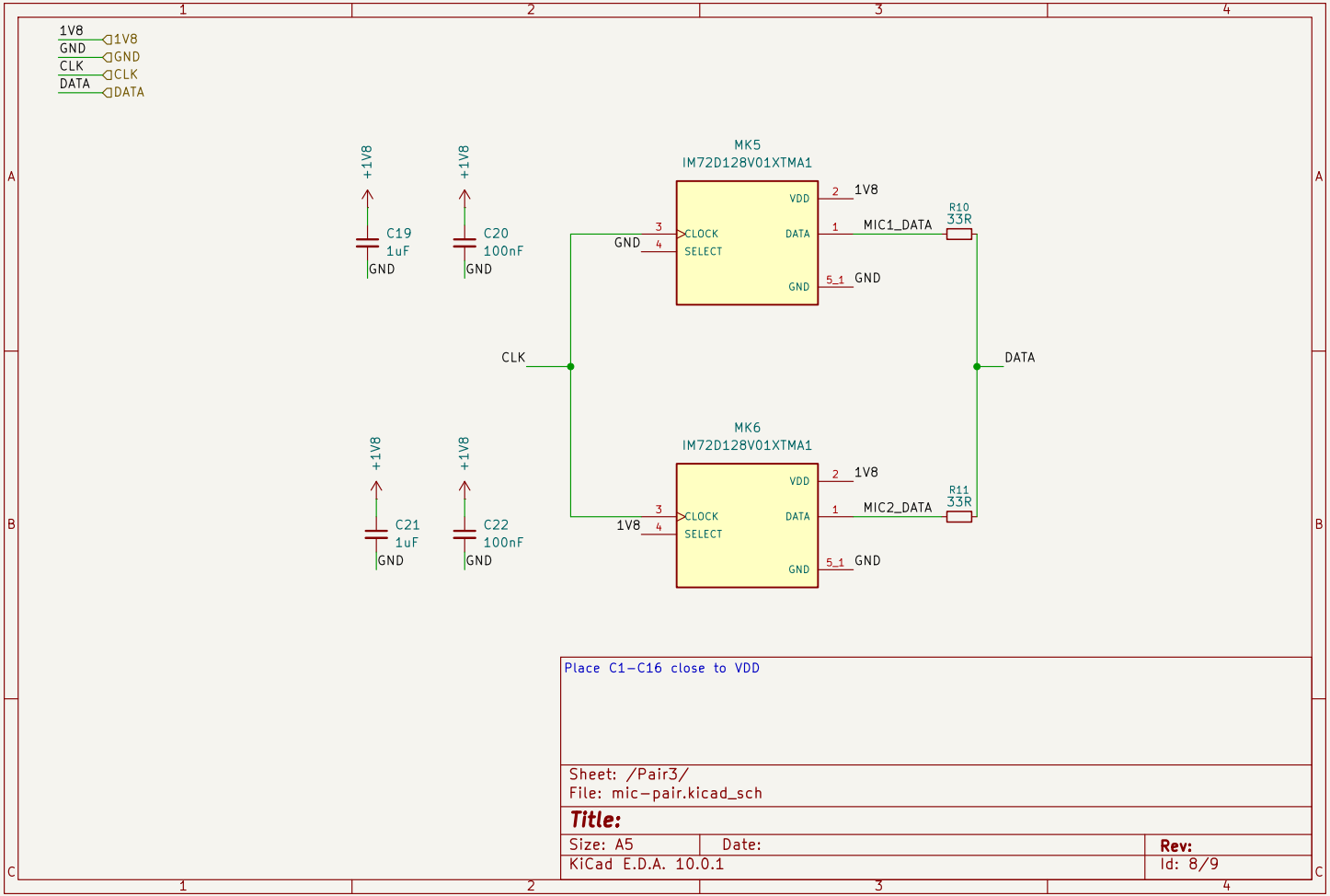

1V8
GND
CLK
DATA
1V8
GND
CLK
DATA
+1V8
C35
1uF
GND
+1V8
C36
100nF
GND
MK13
IM72D128V01XTMA1
VDD
2
1V8
CLOCK
SELECT
3
GND
4
DATA
1
MIC1_DATA
R18
33R
GND
5_1
GND
CLK
DATA
MK14
IM72D128V01XTMA1
VDD
2
1V8
CLOCK
SELECT
3
1V8
4
DATA
1
MIC2_DATA
R19
33R
GND
5_1
GND
+1V8
C37
1uF
GND
+1V8
C38
100nF
GND
Place C1–C16 close to VDD
Sheet: /Pair7/
File: mic–pair.kicad_sch
Title:
Size: A5
Date:
Rev:
KiCad E.D.A. 10.0.1
Id: 3/9

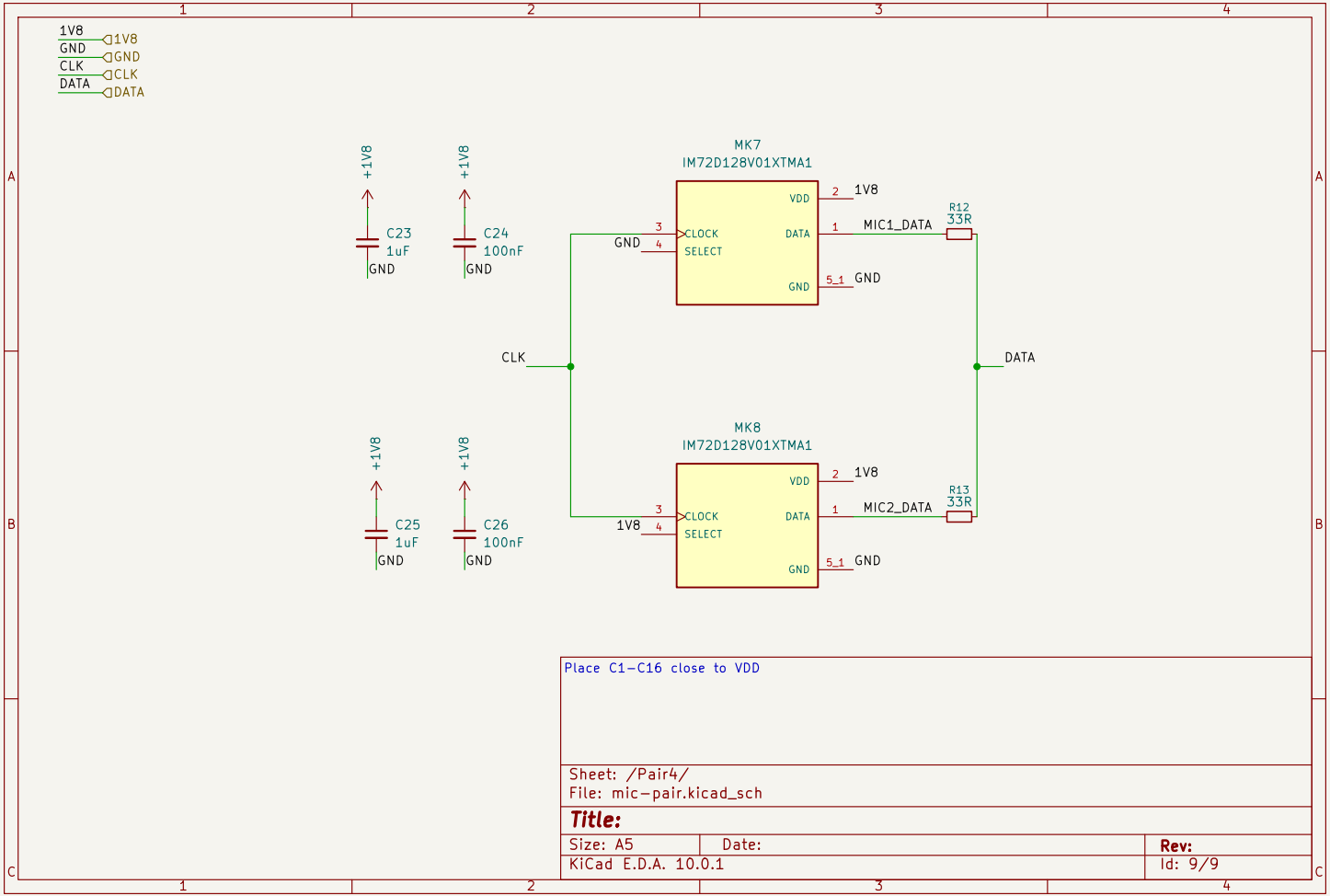

1V8
GND
CLK
DATA
1V8
GND
CLK
DATA
+1V8
C31
1uF
GND
+1V8
C32
100nF
GND
MK11
IM72D128V01XTMA1
VDD
2
1V8
CLOCK
3
GND
SELECT
4
DATA
1
MIC1_DATA
R16
33R
GND
5_1
GND
CLK
DATA
MK12
IM72D128V01XTMA1
VDD
2
1V8
CLOCK
3
1V8
SELECT
4
DATA
1
MIC2_DATA
R17
33R
GND
5_1
GND
+1V8
C33
1uF
GND
+1V8
C34
100nF
GND
Place C1–C16 close to VDD
Sheet: /Pair6/
File: mic–pair.kicad_sch
Title:
Size: A5
Date:
Rev:
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1V8
GND
CLK
DATA
1V8
GND
CLK
DATA
+1V8
C27
1uF
GND
+1V8
C28
100nF
GND
MK9
IM72D128V01XTMA1
VDD
2
1V8
3
CLOCK
GND
4
SELECT
DATA
1
MIC1_DATA
R14
33R
GND
5_1
GND
CLK
DATA
MK10
IM72D128V01XTMA1
VDD
2
1V8
3
CLOCK
1V8
4
SELECT
DATA
1
MIC2_DATA
R15
33R
GND
5_1
GND
+1V8
C29
1uF
GND
+1V8
C30
100nF
GND
Place C1–C16 close to VDD
Sheet: /Pair5/
File: mic–pair.kicad_sch
Title:
Size: A5
Date:
Rev:
KiCad E.D.A. 10.0.1
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1V8
GND
CLK
DATA
1V8
GND
CLK
DATA
+1V8
C11
1uF
GND
+1V8
C12
100nF
GND
MK1
IM72D128V01XTMA1
VDD
2
1V8
CLOCK
SELECT
3
GND
4
DATA
1
MIC1_DATA
R6
33R
GND
5_1
GND
CLK
DATA
MK2
IM72D128V01XTMA1
VDD
2
1V8
CLOCK
SELECT
3
1V8
4
DATA
1
MIC2_DATA
R7
33R
GND
5_1
GND
+1V8
C13
1uF
GND
+1V8
C14
100nF
GND
Place C1–C16 close to VDD
Sheet: /Pair1/
File: mic–pair.kicad_sch
Title:
Size: A5
Date:
Rev:
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1V8
GND
CLK
DATA
1V8
GND
CLK
DATA
+1V8
C15
1uF
GND
+1V8
C16
100nF
GND
MK3
IM72D128V01XTMA1
VDD
2
1V8
R8
33R
3
CLOCK
GND
4
SELECT
DATA
1
MIC1_DATA
GND
5_1
GND
CLK
DATA
+1V8
C17
1uF
GND
+1V8
C18
100nF
GND
MK4
IM72D128V01XTMA1
VDD
2
1V8
R9
33R
3
CLOCK
1V8
4
SELECT
DATA
1
MIC2_DATA
GND
5_1
GND
Place C1–C16 close to VDD
Sheet: /Pair2/
File: mic–pair.kicad_sch
Title:
Size: A5
Date:
Rev:
KiCad E.D.A. 10.0.1
Id: 7/9


1V8
GND
CLK
DATA
1V8
GND
CLK
DATA
+1V8
C19
1uF
GND
+1V8
C20
100nF
GND
MK5
IM72D128V01XTMA1
VDD
2
1V8
CLOCK
3
GND
SELECT
4
DATA
1
MIC1_DATA
R10
33R
GND
5_1
GND
CLK
DATA
MK6
IM72D128V01XTMA1
VDD
2
1V8
CLOCK
3
1V8
SELECT
4
DATA
1
MIC2_DATA
R11
33R
GND
5_1
GND
+1V8
C21
1uF
GND
+1V8
C22
100nF
GND
Place C1–C16 close to VDD
Sheet: /Pair3/
File: mic–pair.kicad_sch
Title:
Size: A5
Date:
Rev:
KiCad E.D.A. 10.0.1
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1V8
GND
CLK
DATA
1V8
GND
CLK
DATA
+1V8
C23
1uF
GND
+1V8
C24
100nF
GND
MK7
IM72D128V01XTMA1
VDD
2
1V8
CLOCK
3
GND
SELECT
4
DATA
1
MIC1_DATA
R12
33R
GND
5_1
GND
CLK
DATA
MK8
IM72D128V01XTMA1
VDD
2
1V8
CLOCK
3
1V8
SELECT
4
DATA
1
MIC2_DATA
R13
33R
GND
5_1
GND
+1V8
C25
1uF
GND
+1V8
C26
100nF
GND
Place C1–C16 close to VDD
Sheet: /Pair4/
File: mic–pair.kicad_sch
Title:
Size: A5
Date:
Rev:
KiCad E.D.A. 10.0.1
Id: 9/9